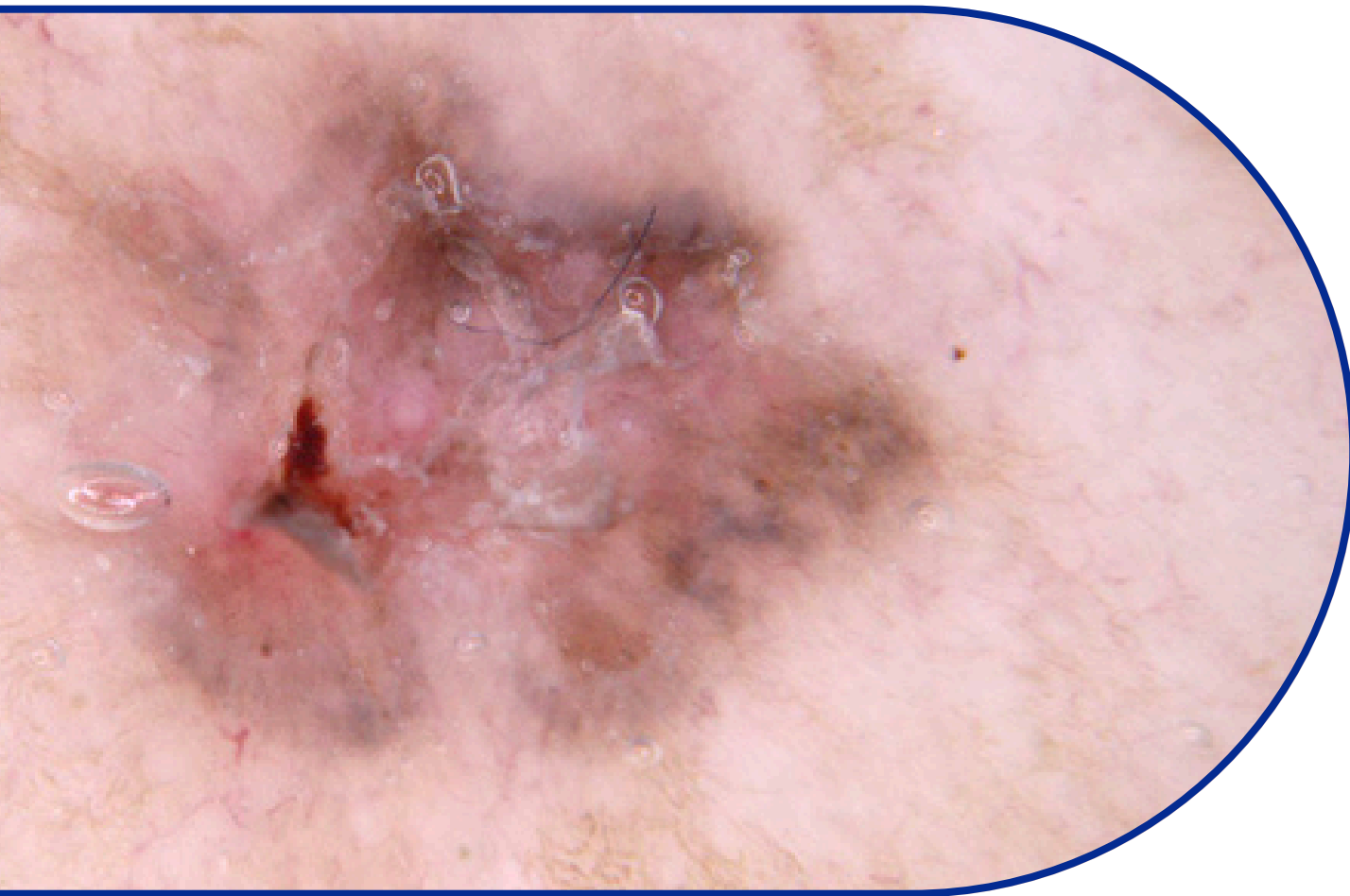


# Skin Cancer Diagnosis



**Magnifying Team** 

## Members

Dinh Xuan Khuong

Nguyen Binh Minh

Nguyen Dang Trung Kien

# Table of contents

**01**

**Problem Statement**

**02**

**Introduction**

**03**

**Core Functionality**

**04**

**Architecture**

**05**

**Estimated Budget**

**06**

**Future Enhancement**

**07**

**Conclusion**

**08**

**Big Thanks**

**09**

**Demo**

01

# Problem Statement

## According to WCRF:

- There were **331,722** new cases of skin cancer in 2022.
- Skin cancer is the **17th** most common cancer worldwide.
- The **14th** common cancer in men.
- The **14th** common cancer in women.

Souce: [Skin Cancer Statistics - WCRF.ORG](https://www.wcrf.org/skin-cancer-statistics)

| Rank | Country               | Number  | ASR/100,000 |
|------|-----------------------|---------|-------------|
|      | World                 | 331,722 | 3.2         |
| 1    | US                    | 101,388 | 16.5        |
| 2    | Germany               | 21,976  | 16.5        |
| 3    | UK                    | 19,712  | 15.3        |
| 4    | Australia             | 16,819  | 37.0        |
| 5    | France (metropolitan) | 15,729  | 13.5        |
| 6    | Italy                 | 13,769  | 12.7        |
| 7    | Russia                | 12,903  | 5.3         |
| 8    | Canada                | 11,383  | 14.5        |
| 9    | Brazil                | 9,676   | 3.3         |
| 10   | China                 | 8,789   | 0.37        |



# Main cause of most skin cancers



**Ultraviolet (UV) Radiation**

# Another big reason is:



*“People do not know they are about to get **skin cancer**”*

# 02

# Introduction

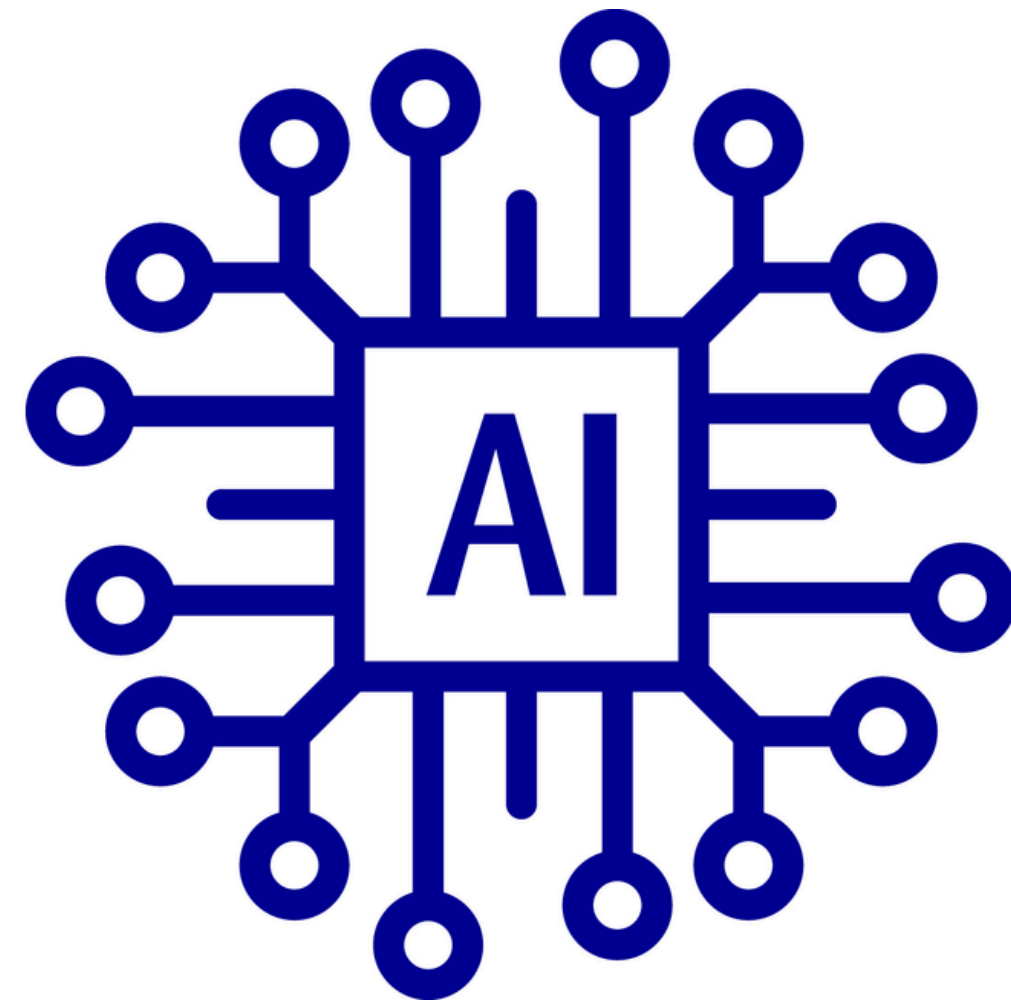


# Say “hello” to: The Skin Cancer Diagnosis (SCD)



UP4000 - edge

+



SCD - Model

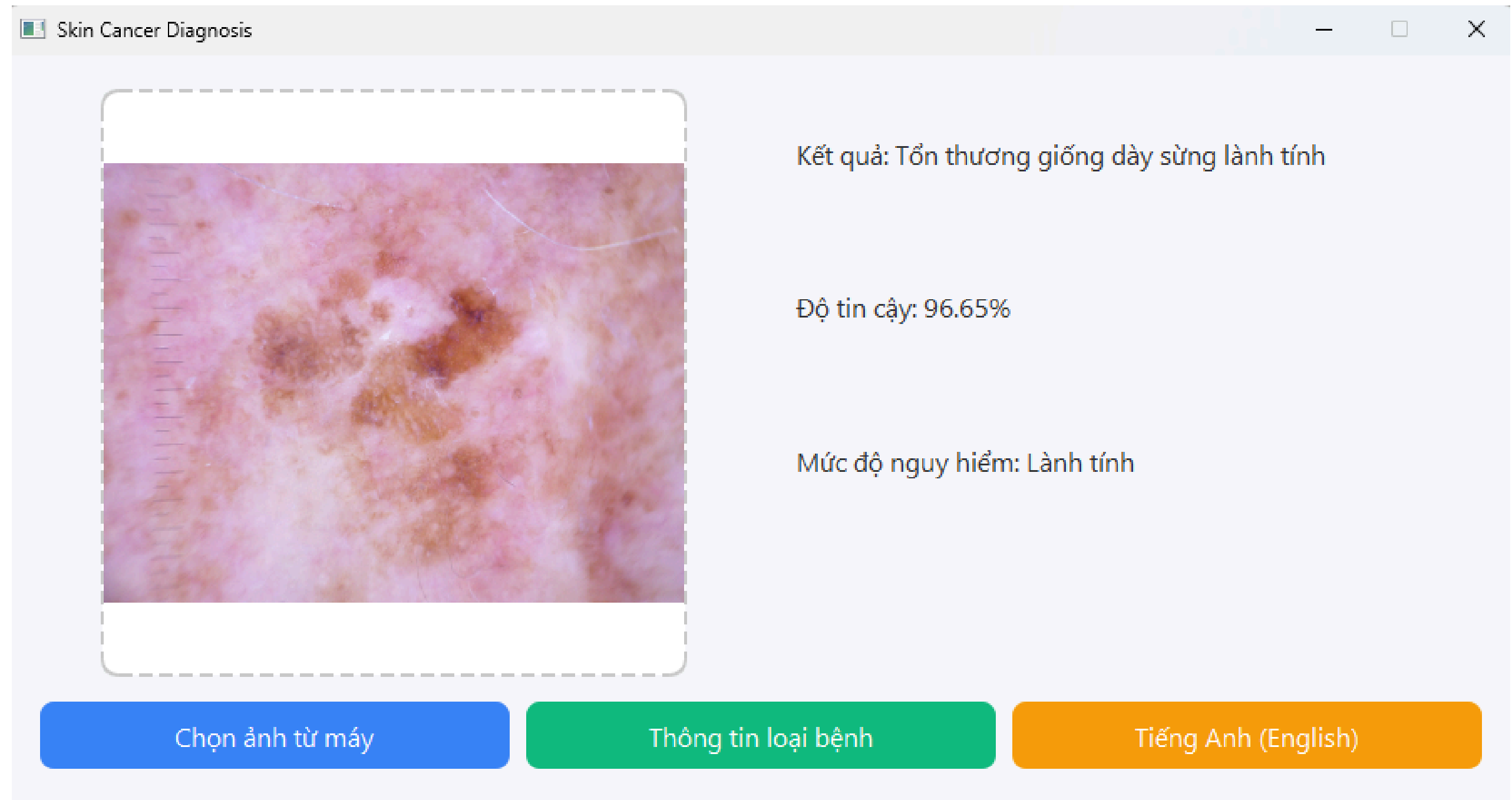
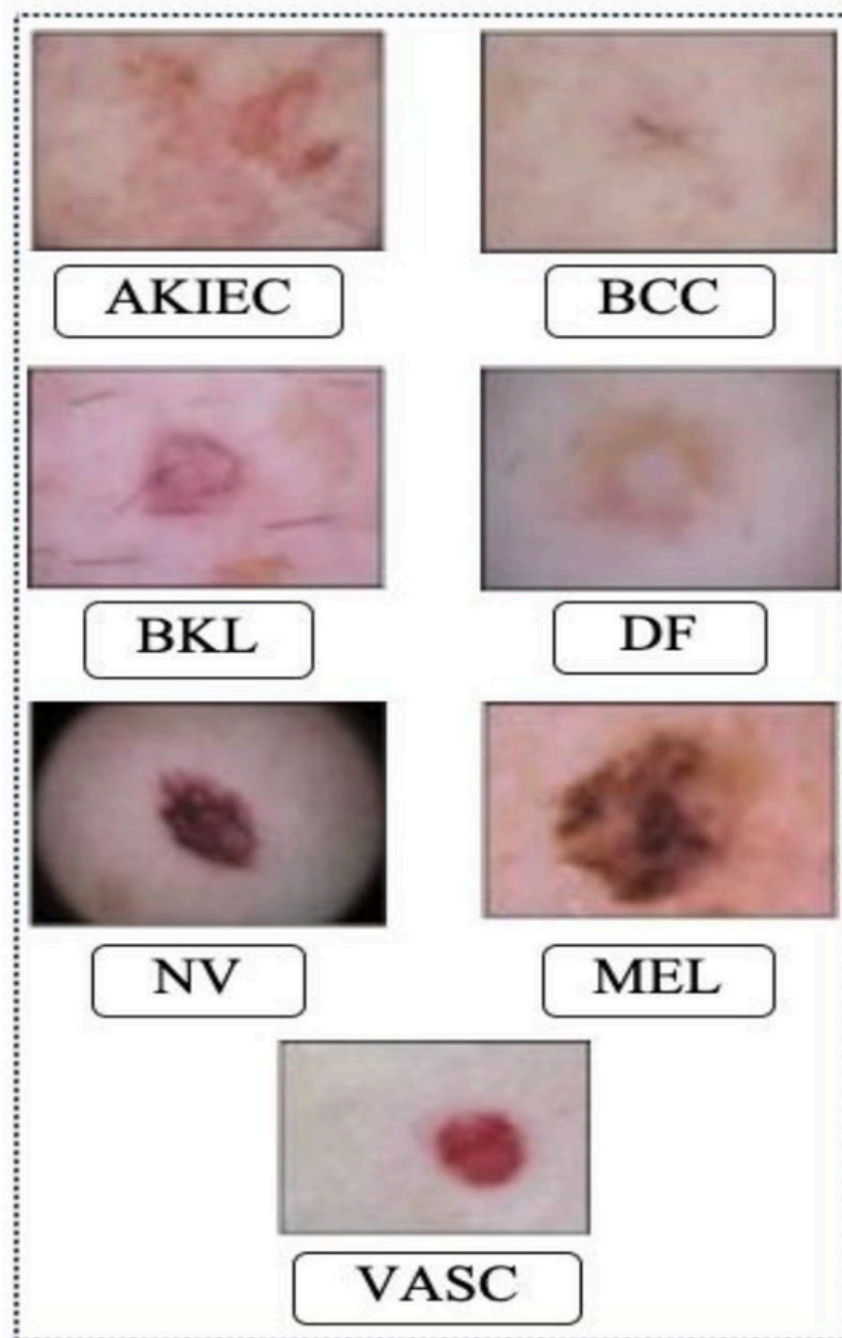
# What is SCD?

- AIoT project
- A deep learning model
- Can assist in the early diagnosis of skin cancer by identifying different types of skin lesions from dermoscopic images.

03

# Core Functionality

# Diagnosing 7 Common Types of Pigmented Skin Lesions



# Disease Information

Thông tin loại bệnh



Thông tin chi tiết các loại bệnh

**BKL:**  
Tổn thương giống dày sừng lành tính  
⚠️ **Mức độ nguy hiểm: Lành tính**

**BCC:**  
Ung thư biểu mô tế bào đáy  
⚠️ **Mức độ nguy hiểm: Ung thư nhẹ**

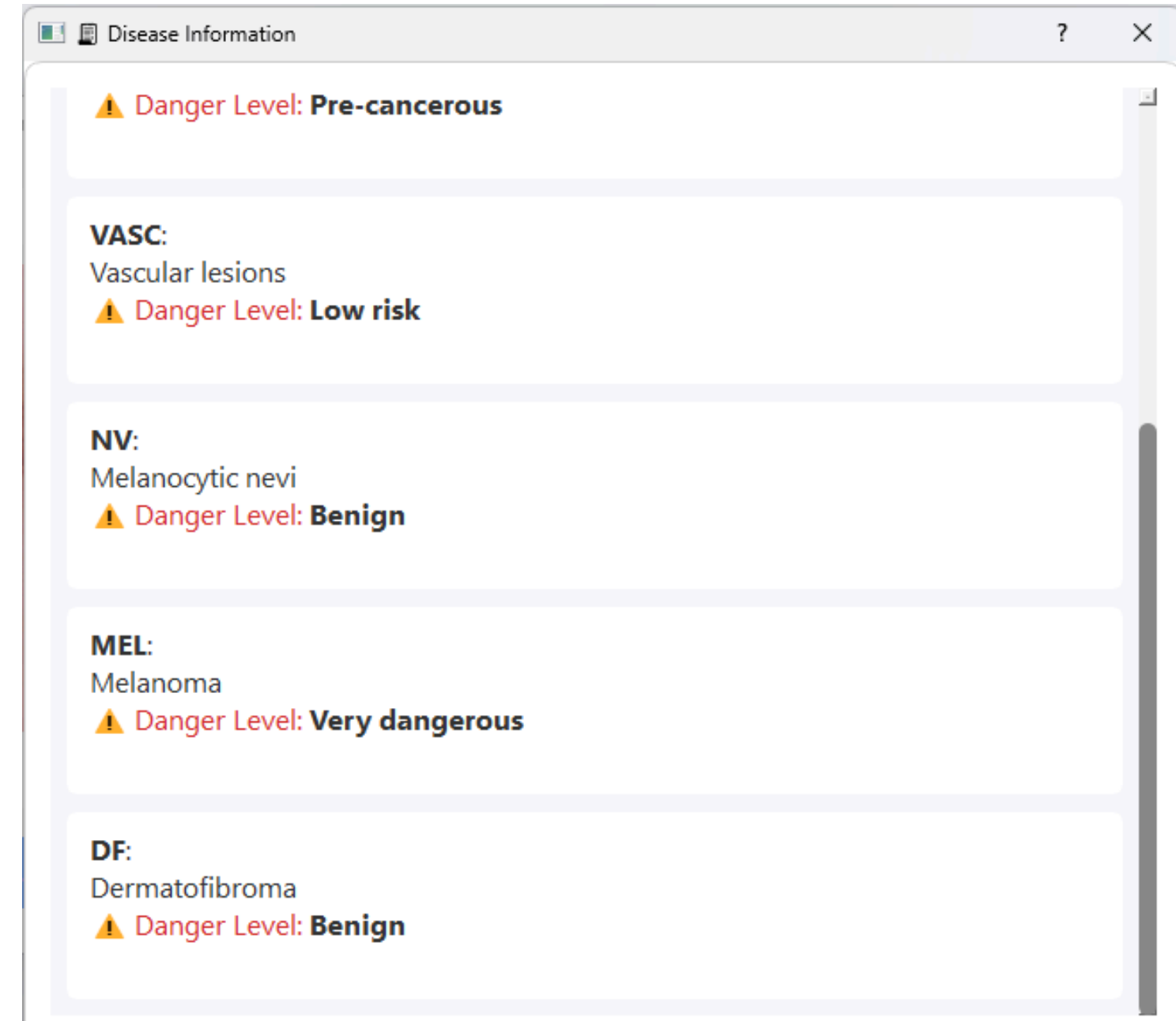
**AKIEC:**  
Dày sừng quang hóa và ung thư biểu mô tại chỗ  
⚠️ **Mức độ nguy hiểm: Tiền ung thư**

**VASC:**  
Tổn thương mạch máu  
⚠️ **Mức độ nguy hiểm: Ít nguy hiểm**

**NV:**  
Nốt ruồi sắc tố



# A user-friendly interface that supports both Vietnamese and English.



## Lightweight SCD

# 04

# Architecture

# Dataset: HAM10000 Harvard

[Add Data](#) ▾[Search](#) ▾[About](#)[User Guide](#)[Support](#)[Sign Up](#)[Log In](#)[ViDIR Dataverse](#)

(Medical University of Vienna)

[Harvard Dataverse](#) > [ViDIR Dataverse](#) >

## The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions

Version 4.0



Tschandl, Philipp, 2018, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions", <https://doi.org/10.7910/DVN/DBW86T>, Harvard Dataverse, V4, UNF:6:KCZFcBLiFE5ObWcTc2ZBOA== [fileUNF]

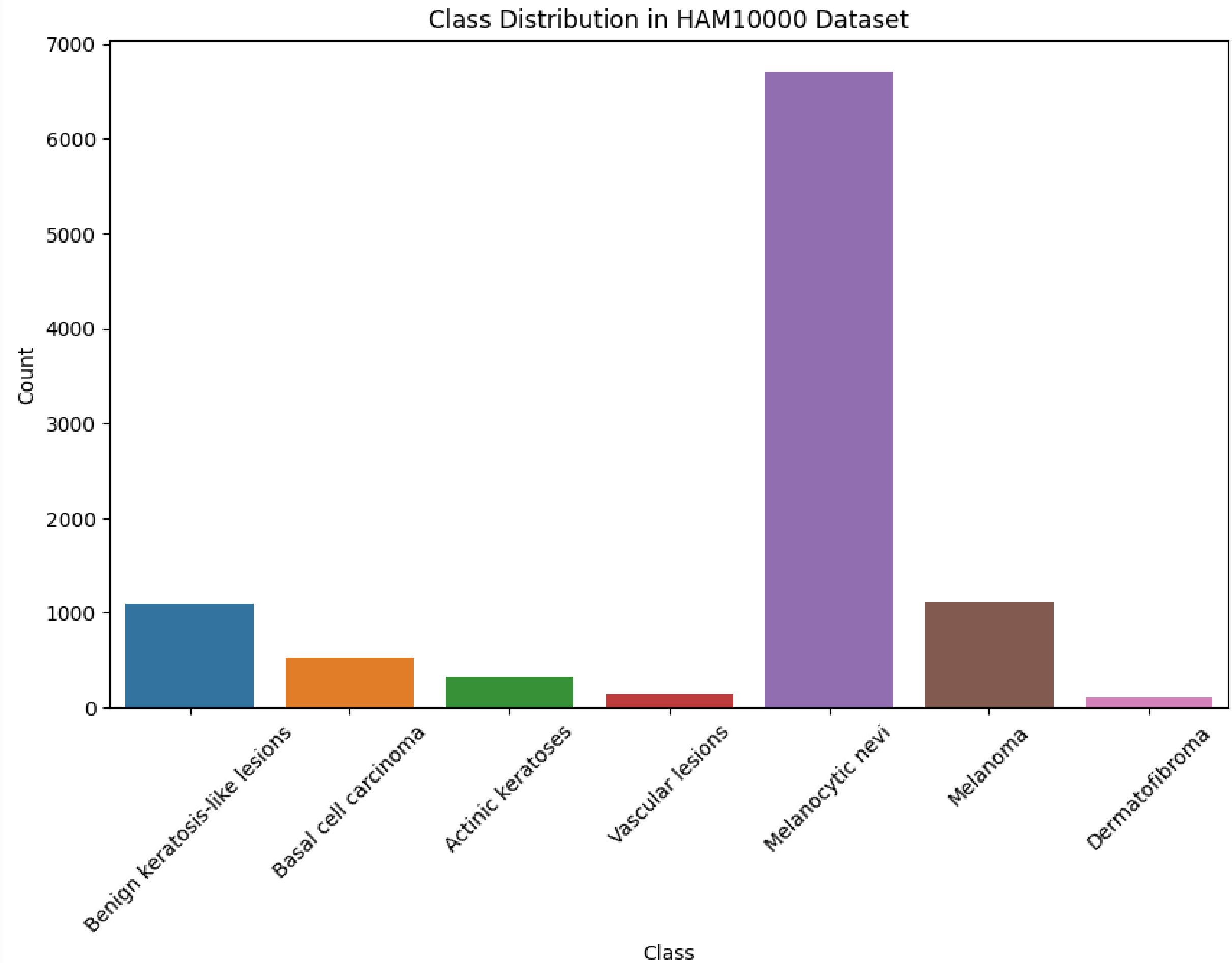
[Cite Dataset](#) ▾[Learn about Data Citation Standards.](#)[Access Dataset](#) ▾[Contact Owner](#)[Share](#)

Dataset Metrics [?](#)

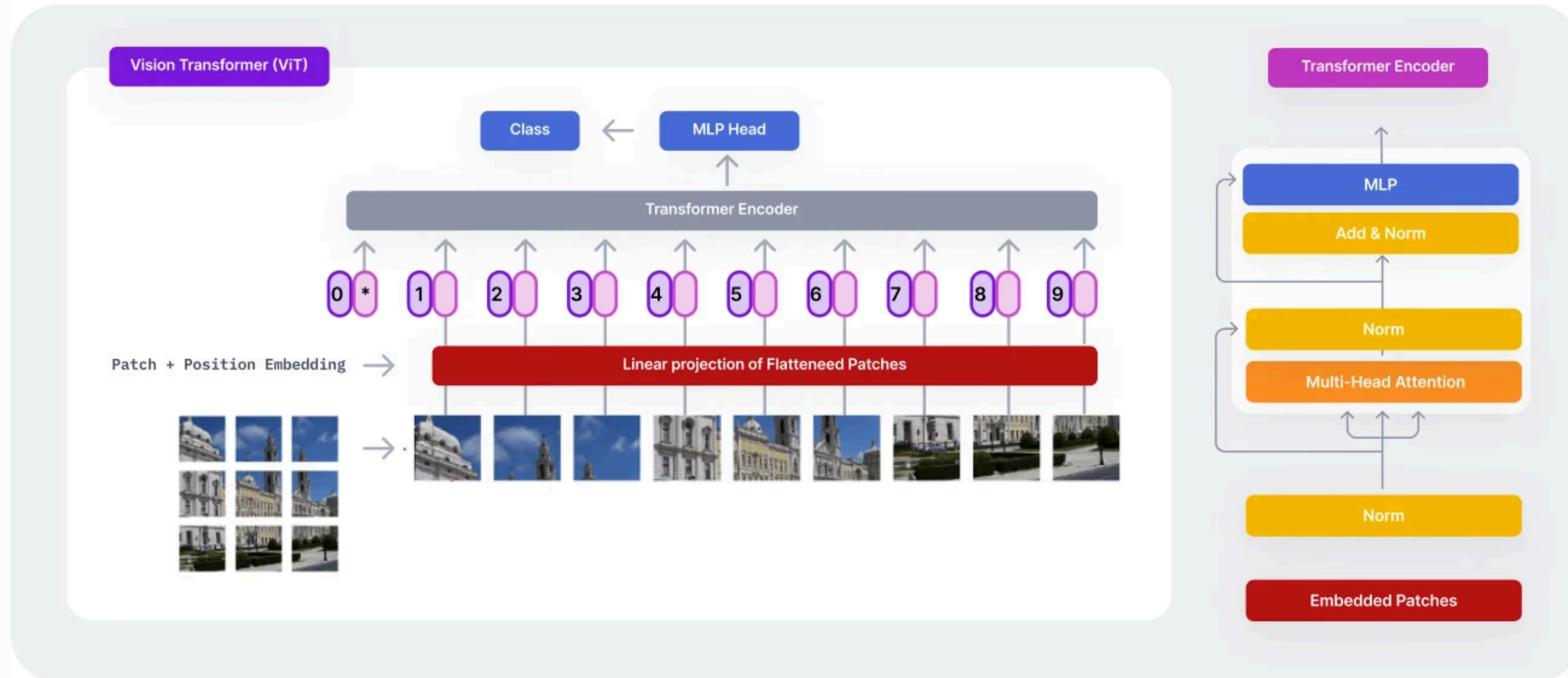
185,830 Downloads [?](#)

# Class Distribution

- 10015 images
- 7 Classes



# Architecture: Vision Transformer



Source: [Vision Transformer: What It Is & How It Works](#)

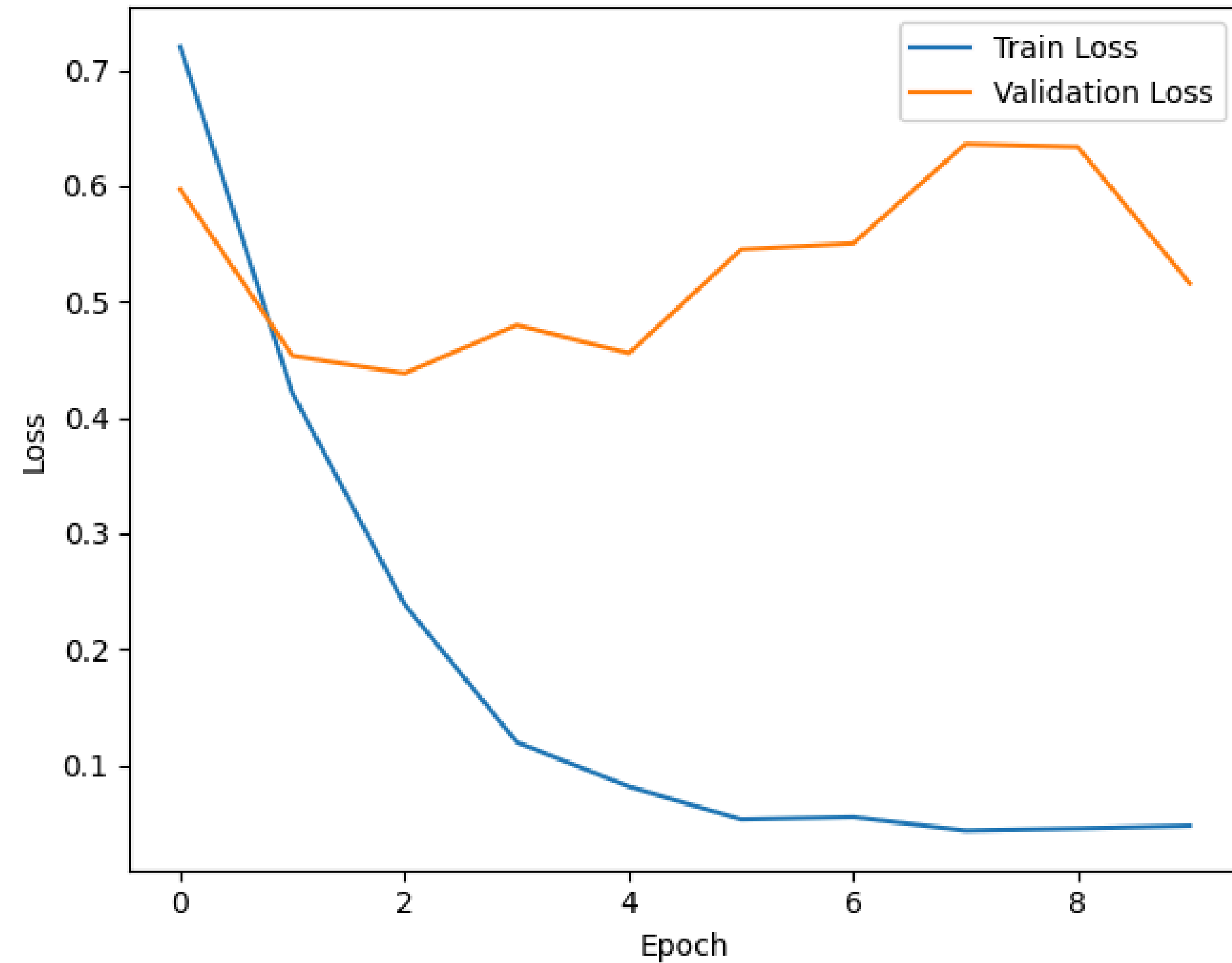


# What did we do?

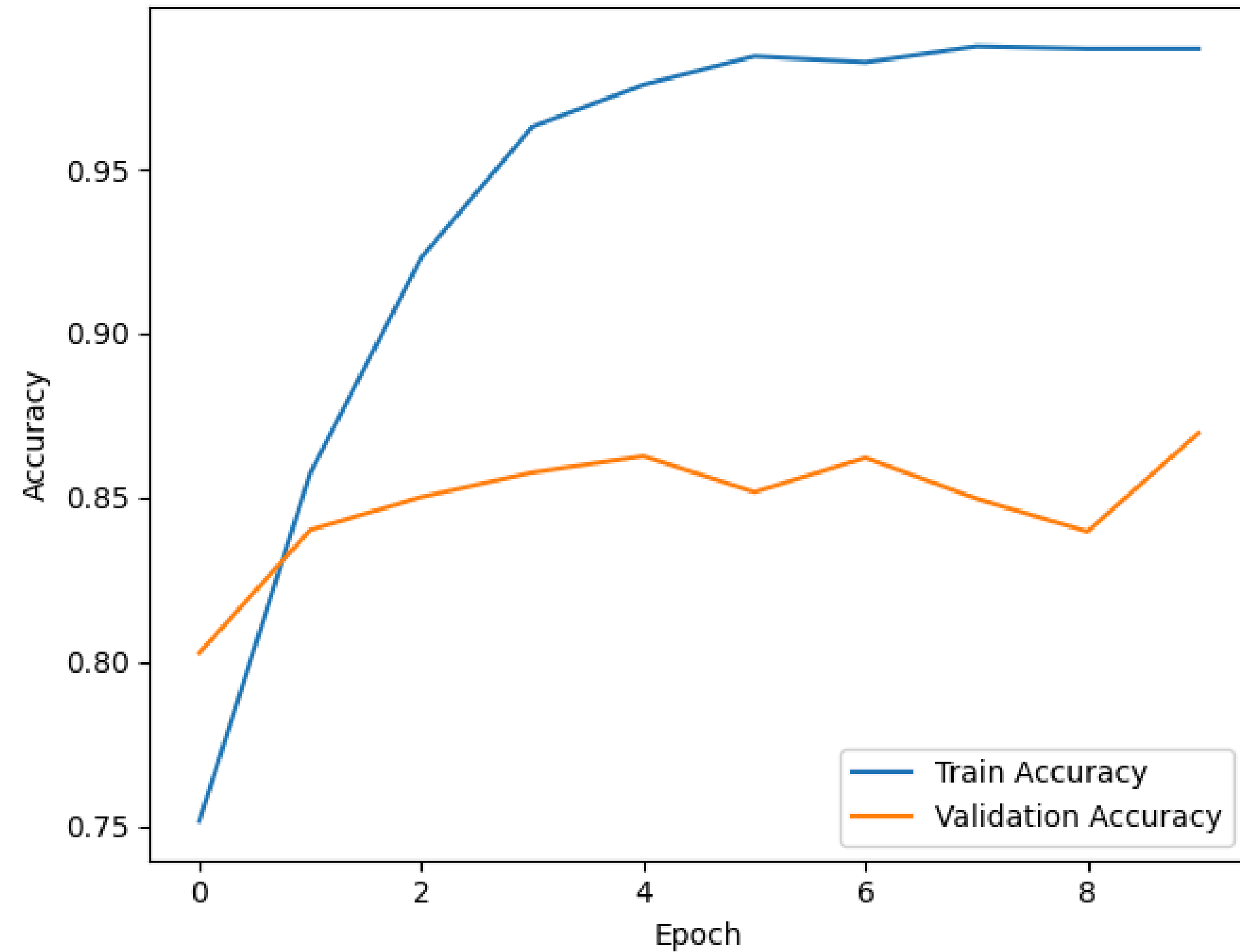
- Use pre-trained ViT model from Hugging Face Transformer.
- Replace the classification heads to match the HAM10000 labels.
- Process data (resize, normalize, ...).
- Fine tune (Cross-entropy loss, Adam optimizer).
- Evaluate the outcome model.

# Training: 10 epochs

Loss Over Epochs

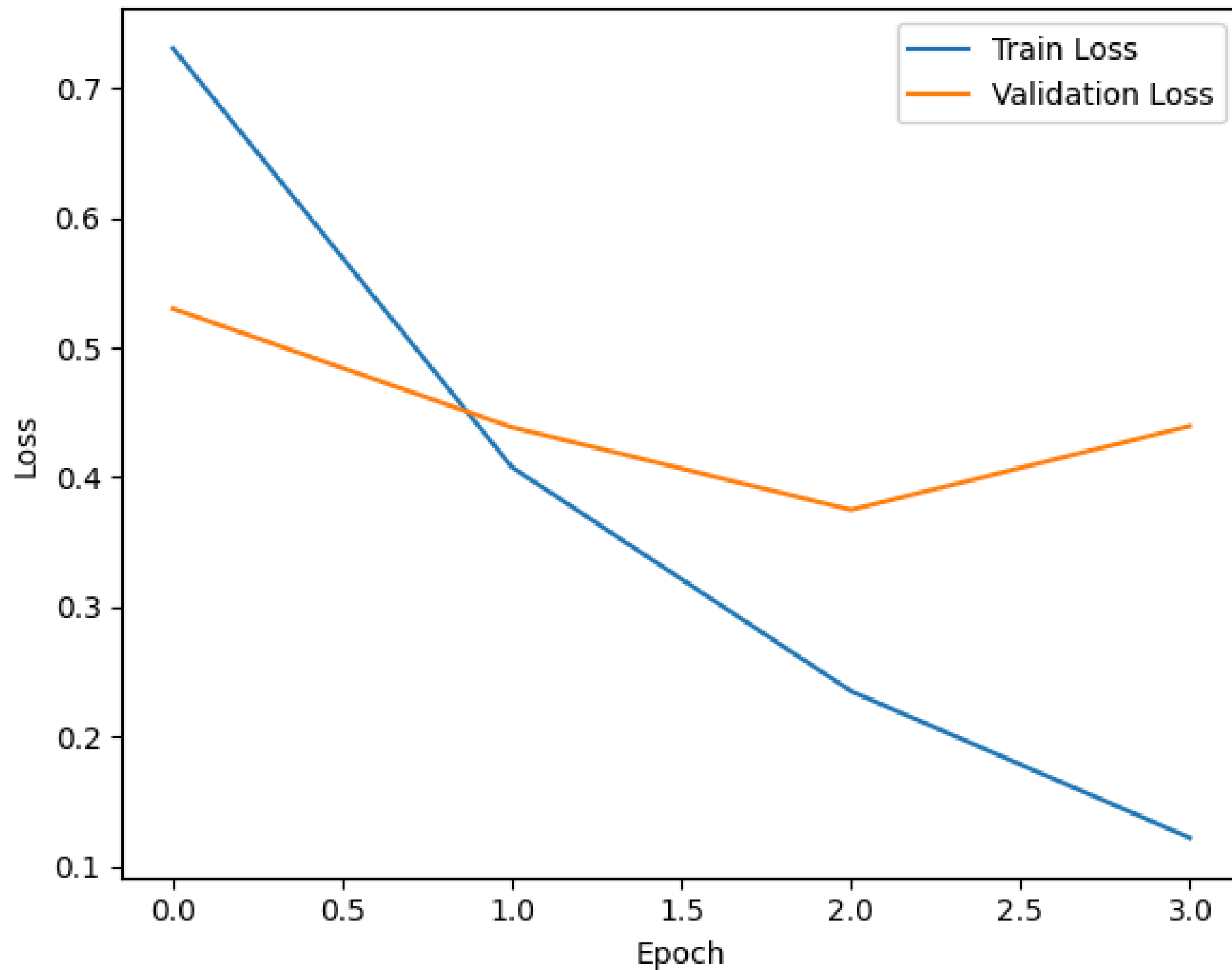


Accuracy Over Epochs

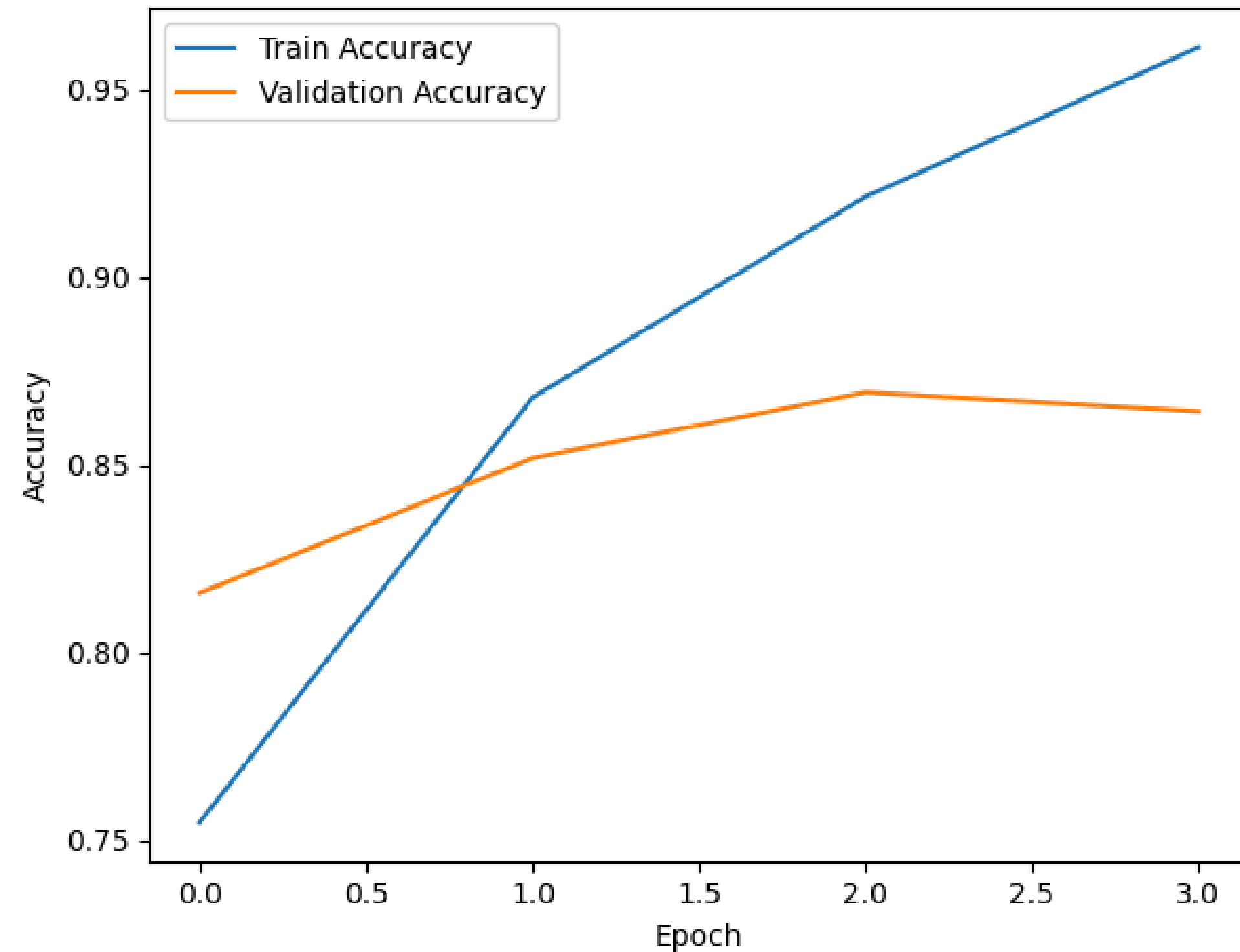


# Training: 4 epochs

Loss Over Epochs



Accuracy Over Epochs

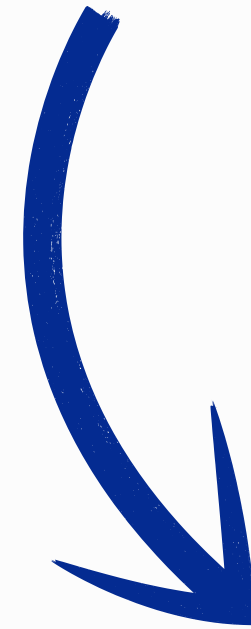




4 epochs



10 epochs



**Choose the model trained with 4 epochs**

**~~Overfitting~~**



# Classification report

## Classification Report:

|                               | precision | recall | f1-score | support |
|-------------------------------|-----------|--------|----------|---------|
| Benign keratosis-like lesions | 0.85      | 0.62   | 0.72     | 220     |
| Basal cell carcinoma          | 0.80      | 0.77   | 0.78     | 103     |
| Actinic keratoses             | 0.49      | 0.83   | 0.62     | 65      |
| Vascular lesions              | 1.00      | 0.89   | 0.94     | 28      |
| Melanocytic nevi              | 0.93      | 0.95   | 0.94     | 1341    |
| Melanoma                      | 0.69      | 0.61   | 0.65     | 223     |
| Dermatofibroma                | 0.60      | 0.91   | 0.72     | 23      |
| accuracy                      |           |        | 0.86     | 2003    |
| macro avg                     | 0.77      | 0.80   | 0.77     | 2003    |
| weighted avg                  | 0.87      | 0.86   | 0.86     | 2003    |

86%!



05

# Estimated Budget

# Estimated Budget

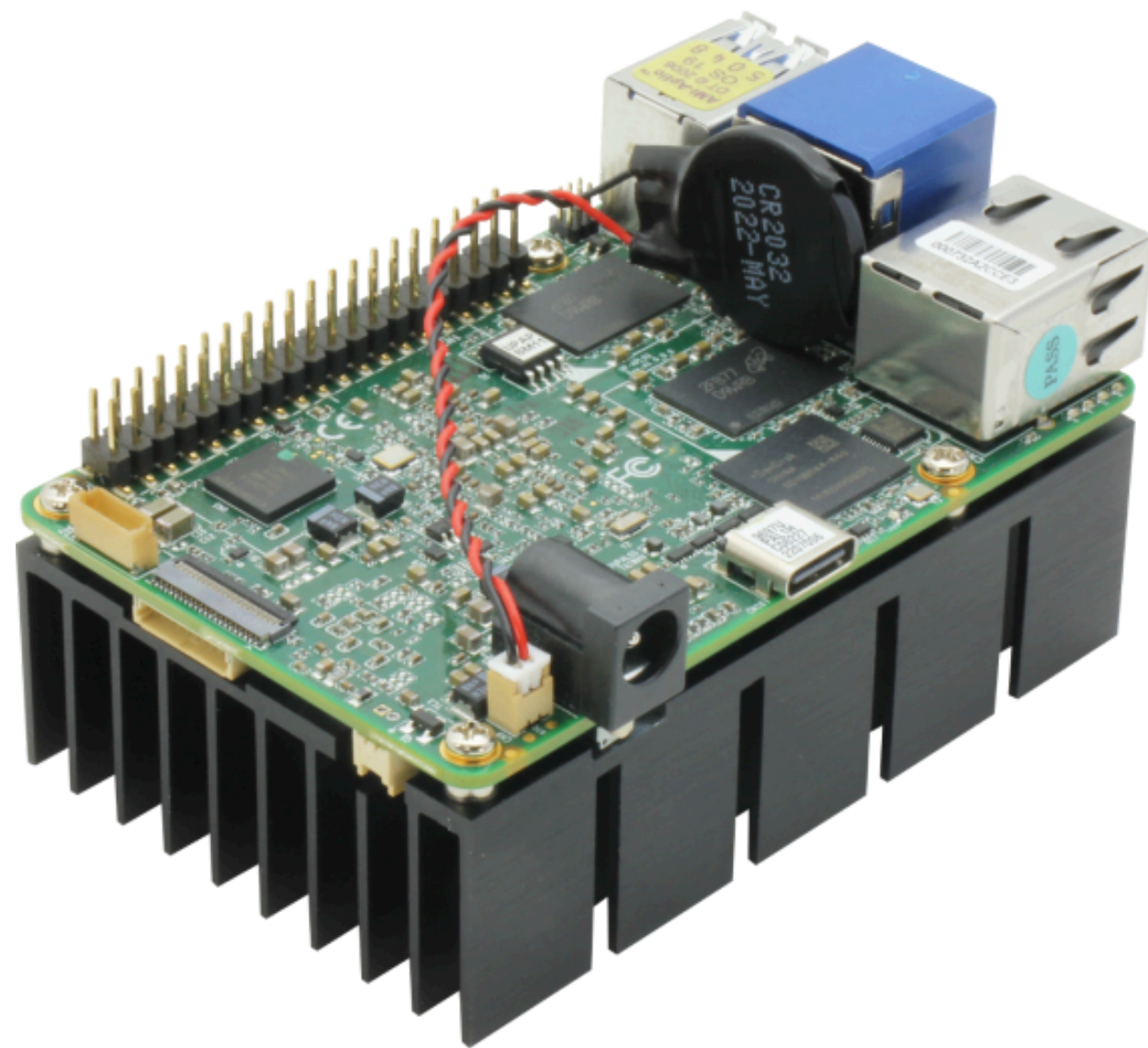
## UP 4000 Celeron 02/32 Low Heatsink

- Processor: Intel® Celeron® N3350
- Graphics: Intel® HD Graphics 500
- Memory: 2GB
- eMMC / Storage: 32 GB

Price: **149\$** (Tax-exclusive)

*It is free for everyone  
if deployed on the web!*

**YEAH!**



# 06

# Future

# Enhancement

# Future Enhancement

- Collecting more data (from hospital, International Skin Imaging Collaboration (ISIC), ...)
- Augmenting data
- Retraining model
- Creating a Multi-language-friendly user interface
- Deploying on the web (free for everyone)

07

# Conclusion



# Conclusion

- This work illustrates how state-of-the-art transformer models **can be adapted to medical domains**, offering **promising support tools** for early and accurate skin cancer detection.
- However, challenges such as class imbalance affected the performance on rare categories
- This suggests future improvements can focus on enhancing data diversity through augmentation or rebalancing techniques.

*Everyone is welcome to contribute to this project!*



DinhXuanKhuong/**Skin-  
Cancer-Diagnosis-with-...**



1

Contributor

0

Issues

0

Stars

0

Forks



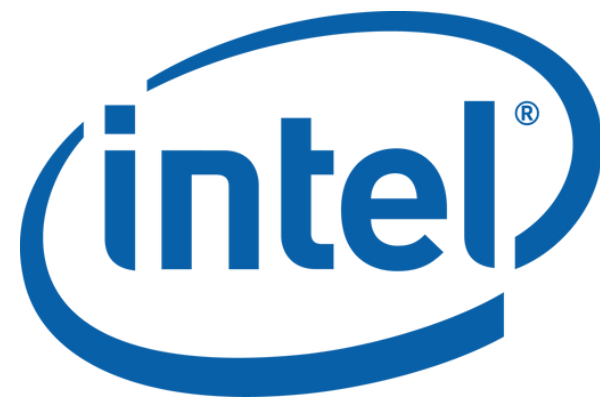
[https://github.com/DinhXuanKhuong/Skin-  
Cancer-Diagnosis-with-Vision-Transformer](https://github.com/DinhXuanKhuong/Skin-Cancer-Diagnosis-with-Vision-Transformer)

**It's open on my github**





# Big thanks!



**Do you have any questions?**

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